

Secure-Tunnel: Post-Quantum Cryptographic Overlay for UAV C2 Links

Results and Analysis Section — v2.0 (2026-03-03)

Draft compilation for review

TABLE I
ALGORITHM ALIASES AND NIST SECURITY LEVELS

Alias	Full Name	Level	Type
ML512	ML-KEM-512	L1	KEM
ML768	ML-KEM-768	L3	KEM
ML1024	ML-KEM-1024	L5	KEM
HQ128	HQC-128	L1	KEM
HQ192	HQC-192	L3	KEM
HQ256	HQC-256	L5	KEM
MC348	Classic-McEliece-348864	L1	KEM
MC460	Classic-McEliece-460896	L3	KEM
MC8192	Classic-McEliece-8192128	L5	KEM
DSA44	ML-DSA-44	L1	SIG
DSA65	ML-DSA-65	L3	SIG
DSA87	ML-DSA-87	L5	SIG
F512	Falcon-512	L1	SIG
F1024	Falcon-1024	L5	SIG
SP128	SPHINCS ⁺ -SHA2-128s	L1	SIG
SP192	SPHINCS ⁺ -SHA2-192s	L3	SIG
SP256	SPHINCS ⁺ -SHA2-256s	L5	SIG
AESG	AES-256-GCM	—	AEAD
CH20	ChaCha20-Poly1305	—	AEAD
ASC	Ascon-128a	—	AEAD

I. RESULTS AND ANALYSIS

All measurements were collected on a Raspberry Pi 4 Model B Rev 1.5 (BCM2711, four Cortex-A72 cores at 1.8 GHz, 4 GB RAM) running Debian 12 with kernel 6.12.47 and the performance CPU governor. Power was sampled via an INA219 current sensor (0.1 Ω shunt) at up to 1 kHz with a documented V_{BUS} gain correction of 1.22. Timing used `time.perf_counter_ns()` with garbage collection disabled. Board-level idle power measured $P_{idle} \approx 3.32$ W (30s capture); reported power values include this baseline unless stated otherwise. The Cortex-A72 does *not* implement ARMv8 Crypto Extensions; AES-GCM therefore executes in software table-lookup mode, while ChaCha20-Poly1305 exploits 128-bit NEON SIMD.

Table I defines the short aliases used throughout. Implementations: liboqs-python 0.14.0 (KEM/SIG), OpenSSL 3.5.4 (AES-256-GCM, ChaCha20-Poly1305), ascon-c v1.2 opt64 via ctypes (Ascon-128a, KAT-verified).

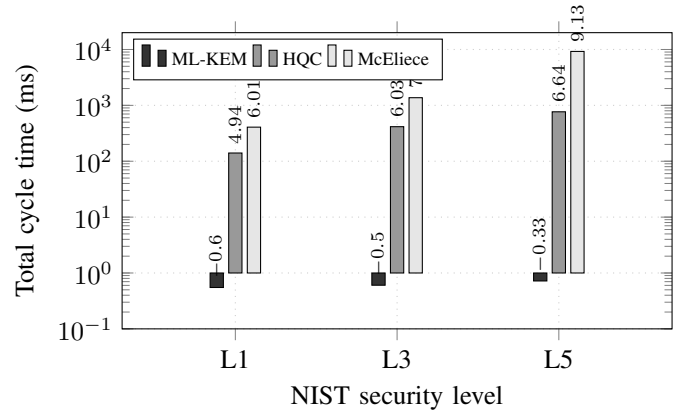


Fig. 1. Total KEM cycle time (keygen+encaps+decaps) by NIST level, log scale. ML-KEM completes in < 1 ms at all levels; HQC and McEliece require 255–12 823 $\times$ longer. 100 iterations, RPi4 @ 1.8 GHz.

A. KEM Primitive Evaluation

Fig. 1 compares total KEM cycle time (keygen+encaps+decaps) across all nine primitives (100 iterations, 1 kHz INA219 sampling).

Table II details all nine KEM primitives. ML-KEM completes a full cycle in under 0.72 ms at Level 5 (0.550 ms at L1, 0.604 ms at L3), drawing 3.46–3.48 W. HQC’s code-based design incurs 255–1 066 $\times$ higher cost, driven by decapsulation (73–393 ms). Classic McEliece exhibits extreme keygen asymmetry: MC8192 requires 9.02 s for keygen but only 1.89 ms for encapsulation, making it practical only with pre-provisioned keys. Only ML-KEM enables sub-second rekey on ARM Cortex-A72.

B. Signature Primitive Evaluation

Table III evaluates eight signature schemes. Keygen is a one-time cost (0.4–282 ms for ML-DSA/Falcon, 187–282 ms for SPHINCS⁺); the per-handshake sign and verify determine rekey latency.

Falcon achieves the lowest combined sign+verify latency at L1 (0.970 ms) and L5 (1.732 ms), outperforming ML-DSA by 36% and 33% respectively, despite 51–73 $\times$ slower keygen (amortized over key lifetime). SPHINCS⁺ imposes prohibitive

TABLE II
KEM PRIMITIVE PERFORMANCE ON RP14 (100 ITER, 1 KHz INA219)

Alias	Lvl	Keygen (ms)	Encaps (ms)	Decaps (ms)	E_{tot} (μJ)	ρ
ML512	L1	0.240	0.152	0.158	1 907	1.00×
HQ128	L1	22.33	44.68	73.39	535 253	255×
MC348	L1	351.0	0.466	55.98	1 737 186	741×
ML768	L3	0.237	0.182	0.185	2 112	1.00×
HQ192	L3	67.68	135.8	211.9	1 651 813	688×
MC460	L3	1 281	0.872	89.59	6 025 434	2 272×
ML1024	L5	0.273	0.216	0.231	2 507	1.00×
HQ256	L5	124.4	250.2	392.6	3 136 867	1 066×
MC8192	L5	9 021	1.894	210.1	40 583 491	12 823×

E_{tot} : total energy (keygen+encaps+decaps), INA219-measured. ρ : total cycle time relative to ML-KEM at same NIST level.

TABLE III
SIGNATURE SIGN+VERIFY ON RP14 (100 ITER, 1 KHz INA219)

Alias	Lvl	Sign (ms)	Verify (ms)	E_s (μJ)	E_v (μJ)	ρ
DSA44	L1	1.179	0.331	4 264	1 175	1.00×
F512	L1	0.778	0.192	2 773	683	0.64×
SP128	L1	1 468	1.582	6 008 719	5 690	972×
DSA65	L3	1.633	0.484	5 849	1 723	1.00×
SP192	L3	2 610	2.299	10 628 558	8 318	1 234×
DSA87	L5	1.847	0.723	6 612	2 564	1.00×
F1024	L5	1.452	0.280	5 175	995	0.67×
SP256	L5	2 321	3.292	9 516 164	11 738	904×

ρ : sign+verify cost vs. ML-DSA at same level. Falcon $\rho < 1$ = faster combined sign+verify. SP128 sign = 1.47 s · 6.01 J per invocation.

signing latency: SP128 at 1.47 s (6.01 J per signature) and SP192 at 2.61 s (10.63 J), approaching the rekey period at short intervals. Verification remains fast (≤ 3.3 ms) for all algorithms, confirming the GCS verification path is never the bottleneck. Falcon is therefore the preferred signature scheme where its non-standard keygen model is acceptable; ML-DSA provides the general-purpose default at under 2.6 ms combined cost through Level 5.

C. AEAD Primitive Evaluation

AEAD selection operates exclusively on the UDP data plane and does *not* affect handshake timing. Table IV reports results from 10 000 iterations per configuration (500-iteration warmup, two-pass methodology: timing pass with GC disabled, power pass with INA219 at ~ 103 Hz).

Timing distributions are concentrated: $\sigma_{\text{enc}} = 0.38 \mu\text{s}$ (AESG) and $0.77 \mu\text{s}$ (ASC). CH20 exhibits $\sigma = 3.99 \mu\text{s}$ due to rare OS scheduling jitter (P95 = 6.46, P99 = 6.69 μs over 10 000 samples; max = 208 μs), with median 6.30 μs confirming the mean is representative.

Table V and Fig. 2 show scaling across payload sizes spanning MAVLink telemetry (9–263 B) to bulk transfer (4096 B).

At 4096 B the CH20:AESG throughput ratio reaches 3.93 $\times$, while below 64 B setup cost dominates and the ciphers converge. The energy-per-bit metric at 256 B yields

TABLE IV
AEAD PERFORMANCE AT 256-BYTE PAYLOAD (10 000 ITERATIONS, RP14)

Alias	Enc (μs)	Dec (μs)	P_{enc} (W)	P_{dec} (W)	$E_{256\text{B}}$ (μJ)	E_{bit} (nJ/bit)
AESG	9.25	9.09	3.63	3.57	66.0	32.2
CH20	6.47	6.05	3.54	3.58	44.6	21.8
ASC	12.72	12.85	3.84	3.59	95.0	46.4

$E_{256\text{B}} = P_{\text{avg}} \times (t_{\text{enc}} + t_{\text{dec}})$. $E_{\text{bit}} = E_{256\text{B}}/2048$. Board-level power; $P_{\text{idle}} = 3.320$ W not subtracted. Crypto-only ΔP : AESG 0.31, CH20 0.22, ASC 0.52 W above idle.

TABLE V
AEAD ENCRYPT LATENCY SCALING BY PAYLOAD SIZE (μs)

Alias	9 B	33 B	64 B	256 B	1024 B	4096 B
AESG	4.75	5.45	5.96	9.25	22.98	75.71
CH20	5.30	5.38	5.33	6.47	9.38	19.25
ASC	11.38	11.82	11.78	12.72	17.27	30.86

Per-KiB slopes: CH20 3.4, ASC 4.8, AESG 17.7 $\mu\text{s}/\text{KiB}$.

CH20 = 21.8 nJ/bit, AESG = 32.2 nJ/bit, ASC = 46.4 nJ/bit. For a typical 50 Hz MAVLink stream (263 B packets), per-second AEAD cost ranges from 2.23 mJ (CH20) to 4.75 mJ (ASC)—negligible versus the 3.32 W board draw. ChaCha20-Poly1305 is therefore the optimal data-plane cipher for ARM cores without hardware AES acceleration, delivering 32% lower energy per bit than AES-256-GCM.

D. Suite-Level Evaluation

Each of the 24 level-matched KEM $\times$ SIG combinations was benchmarked end-to-end on the tunnel prototype (10 s per suite, baseline with no DDoS detector). T_{HS} is measured from handshake initiation to session establishment, including protocol framing, HKDF key derivation, and Python interpreter overhead. T_{crypto} is the sum of the in-suite KEM and SIG primitive times recorded during the same run. Table VI reports all 24 suites grouped by NIST level (9 at L1, 6 at L3, 9 at L5).

The default suite (ML768+DSA65) completes a full handshake in 12.71 ms at Level 3, of which only 2.71 ms (21%) is cryptographic computation; the remaining ~ 10 ms is protocol framing, HKDF key derivation, and interpreter overhead. At Level 5, ML1024+F1024 measures 8.75 ms—the fastest of all 24 suites—because Falcon-1024’s combined sign+verify (1.18 ms) undercuts DSA65 (1.94 ms) and the crypto share (1.82 ms) is minimal.

Three structural observations emerge from the full 24-suite matrix: (1) ML-KEM suites are the only family achieving sub-100 ms handshakes at every security level ($\rho \leq 1.9\times$); (2) SPHINCS⁺ renders any suite non-viable regardless of KEM choice, with T_{HS} exceeding 655 ms even paired with ML-KEM; and (3) McEliece suites at L5 (MC8192) exceed 500 ms due to keygen asymmetry, making them impractical for real-time rekey. The MC460+SP192 combination timed out at 48.2 s, confirming that the MDEAS scheduler’s cross-axis con-

TABLE VI
MEASURED SUITE HANDSHAKE PERFORMANCE — ALL 24 LEVEL-MATCHED KEM×SIG COMBINATIONS (BASELINE, NO DDoS)

KEM	SIG	T_{HS} (ms)	T_{crypto} (ms)	$\hat{E}_{\text{HS}}$ (mJ)	Crypto (%)	ρ	Viable?
<i>NIST Level 1</i> (3 KEMs × 3 SIGs = 9 suites)							
ML512	F512	12.80	5.69	44.3	44	1.0×	✓
ML512	DSA44	14.34	2.45	49.6	17	1.1×	✓
HQ128	F512	57.80	54.30	200	94	4.5×	✓
HQ128	DSA44	61.43	56.50	213	92	4.8×	✓
MC348	DSA44	187.5	132.3	649	71	14.6×	✓*
MC348	F512	206.1	161.4	713	78	16.1×	✓*
ML512	SP128	655.1	644.8	2 268	98	51.2×	×
HQ128	SP128	722.8	718.1	2 502	99	56.5×	×
MC348	SP128	1 080	1 045	3 739	97	84.3×	×
<i>NIST Level 3</i> (3 KEMs × 2 SIGs = 6 suites)							
ML768	DSA65	12.71	2.71	44.3	21	1.0×	✓
HQ192	DSA65	155.8	159.9	543	100 [‡]	12.3×	✓*
MC460	DSA65	257.4	185.4	897	72	20.2×	✓*
ML768	SP192	1 250	1 233	4 362	99	98.3×	×
HQ192	SP192	1 537	1 533	5 364	100	120.9×	×
MC460	SP192	48 186 [†]	—	—	—	—	×
<i>NIST Level 5</i> (3 KEMs × 3 SIGs = 9 suites)							
ML1024	F1024	8.75	1.82	30.5	21	1.0×	✓
ML1024	DSA87	16.78	2.25	58.5	13	1.9×	✓
HQ256	F1024	271.7	283.7	947	100 [‡]	31.1×	✓*
HQ256	DSA87	276.1	288.9	963	100 [‡]	31.6×	✓*
MC8192	DSA87	505.3	367.3	1 763	73	57.7×	×
MC8192	F1024	678.4	527.9	2 367	78	77.5×	×
ML1024	SP256	1 110	1 092	3 873	98	126.9×	×
HQ256	SP256	1 470	1 494	5 128	100 [‡]	168.0×	×
MC8192	SP256	1 651	1 525	5 759	92	188.7×	×

T_{HS} : measured end-to-end handshake time (protocol + crypto + overhead). T_{crypto} : sum of in-suite KEM and SIG primitive times. $\hat{E}_{\text{HS}}$: estimated energy ($P_{\text{avg}} \times T_{\text{HS}}$, using per-primitive P_{avg} from Tables II–III). ρ : relative to fastest suite at same NIST level. Viable: ✓ real-time compatible ($T_{\text{HS}} < 100$ ms); ✓* viable with extended rekey interval; × non-viable ($T_{\text{HS}} > 500$ ms or timeout). [†]Handshake timeout (48.2 s); crypto fields unavailable. [‡] $T_{\text{crypto}} > T_{\text{HS}}$: measurement jitter across run boundaries; crypto share capped at 100%.

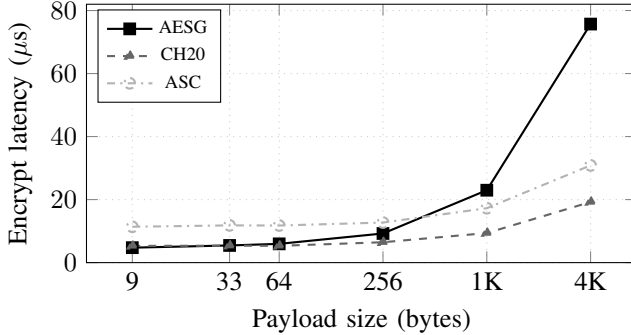


Fig. 2. AEAD encrypt latency versus payload size. CH20 (NEON SIMD) scales at 3.4 μ s/KiB; AESG at 17.7 μ s/KiB (software table-lookup). Below 64 B, setup cost dominates and ciphers converge.

straint forbidding McEliece×SPHINCS⁺ pairs is empirically justified.

E. DDoS–Cryptographic Interaction

Concurrent DDoS detection competes with the cryptographic pipeline for CPU and thermal headroom. All 72 level-matched suites were benchmarked under baseline (no

TABLE VII
PQC HANDSHAKE OVERHEAD UNDER CONCURRENT DETECTION (72 SUITES)

Statistic	XGBoost	TST
Mean ΔT_{HS} (%)	+2.5	+71.8
Median ΔT_{HS} (%)	+0.05	+32.6
Max ΔT_{HS} (%)	+137.2	+405.1
<i>Default suite: ML768+AESG+DSA65</i>		
System-level T_{base} (ms)	13.01	
T_{HS} + detector (ms)	13.46	37.90
ΔT_{HS} (%)	+3.5	+191.3

$\Delta T_{\text{HS}} = (T_{\text{det}} - T_{\text{base}})/T_{\text{base}} \times 100$. T_{base} is the measured system-level handshake (cf. Table VI).

detector), XGBoost, and TST scenarios (10 s per suite per scenario, crypto-only handshakes).¹

XGBoost adds a median 0.05% overhead (mean 2.5%) to PQC handshakes, confirming tree-based detection is compatible with real-time cryptographic operations. TST nearly triples the default suite latency (13.01 $\rightarrow$ 37.90 ms, +191%), driven by

¹Both lightweight detectors use 2 features (Mavlink_Count, Total_length), distinct from the 54-feature CIC-IoT-2023 models in Table VIII. TST uses seq_len=1; functionally a 3-layer MLP with 1.6 M parameters.

TABLE VIII
DDoS DETECTOR INFERENCE ON RPi4 (CIC-IoT-2023)

Model	Acc. (%)	F1	t_{inf} (ms)	P99 (ms)	ΔP (mW)	E_{inf} (mJ)
LightGBM	93.47 [†]	0.937 [†]	6.72	8.59	+999	29.0
XGBoost	94.55 [†]	0.947 [†]	7.42	9.52	+951	31.7
RF	93.35 [†]	0.935 [†]	74.68	102.63	+594	291.4
TST	90.27 [†]	0.80 [†]	10.72	26.12	+1973	56.7

[†] Accuracy and F1 from model documentation; inference metrics measured on RPi4 (15 s per model, idle baseline 3.309 W). CIC-IoT-2023: 3.4 M samples. XGBoost/LightGBM/RF: 54 features, 15 classes. TST: 46 features, 6 classes.

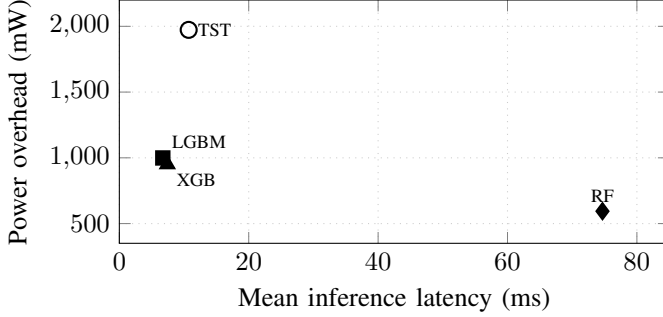


Fig. 3. Detector resource footprint: inference latency vs. power overhead. XGBoost and LightGBM cluster in the low-latency, moderate-power region. RF is a latency outlier (10 $\times$); TST doubles power draw at modest latency.

its 93% CPU utilization starving the handshake of execution time.

Table VIII and Fig. 3 compare the four trained CIC-IoT-2023 detectors on inference cost.

XGBoost achieves the highest reported accuracy (94.55%) at 7.42 ms mean latency and 31.7 mJ per inference. Random-Forest consumes 10 $\times$ more energy (291.4 mJ) due to its 200-tree ensemble. TST delivers the lowest accuracy (90.27%) at the highest power cost (+1973 mW, 93% CPU), with a P99 tail of 26.12 ms indicating variable inference latency. Tree-based models are therefore the preferred architecture for embedded PQC-DDoS co-deployment: XGBoost adds negligible handshake overhead while providing the highest detection accuracy.

F. Scheduler Behavioral Impact

The Multi-Dimensional Energy-Aware Scheduler (MDEAS) manages three decision axes evaluated at 1 Hz: (1) AEAD cipher selection based on energy and thermal state; (2) security-level promotion/demotion triggering PQC rekey; and (3) DDoS detector activation based on threat telemetry and thermal headroom.

Each security-level transition requires a complete PQC handshake. The rekey overhead fraction is:

$$\Phi(R) = \frac{T_{\text{HS}}}{R + T_{\text{HS}}} \quad (1)$$

where R is the rekey interval. For the default suite ($T_{\text{HS}} = 12.71$ ms, Table VI) at the minimum 30 s interval: $\Phi = 4.2 \times 10^{-4}$ (<0.05%). Even the heaviest measured suite

TABLE IX
MDEAS DETECTOR OVERHEAD PARAMETERS

Detector	ΔP (W)	ΔT ($^{\circ}\text{C}$)	ΔCPU (pp)	T_{max} ($^{\circ}\text{C}$)
None	0.00	0.0	0	80.0
XGBoost	0.95	4.8	35	75.0
TST	1.97	10.7	91	69.0

T_{max} : baseline temperature ceiling before detector activation is blocked (80°C throttle minus ΔT). ΔCPU in percentage points.

(ML768+SP192, $T_{\text{HS}} = 1.25$ s) at 30 s yields $\Phi = 0.040$ (4.0%), confirming that rekey overhead is negligible for ML-KEM configurations.

Table IX summarizes the per-detector resource profiles used for proactive thermal management.

The MDEAS enforces cross-axis constraints to prevent pathological configurations: Ascon-128a is forbidden when TST is active, as its higher per-packet cost combined with 93% CPU utilization exceeds the real-time budget; McEliece KEMs are discouraged under TST due to CPU-bound key-gen contention; all McEliece $\times$ SPHINCS⁺ pairs are forbidden because T_{HS} would exceed the 45 s rekey timeout; and detector activation is blocked when temperature exceeds the per-detector T_{max} ceiling. These constraints reduce the 648 theoretical configurations (216 suites $\times$ 3 detector states) to approximately 500 viable operating points, navigating the security–energy–thermal Pareto surface via 1 Hz telemetry feedback while guaranteeing the configured minimum security level.